


```

lubuntu@lubuntu:~$ id qa2
uid=1004(qa2) gid=1004(qa2) groups=1004(qa2),1006(quality)
lubuntu@lubuntu:~$ sudo usermod -g quality qa2
lubuntu@lubuntu:~$ id qa2
uid=1004(qa2) gid=1006(quality) groups=1006(quality)

```

8. Remove `qa1` from the `quality` group without deleting the user.

```

lubuntu@lubuntu:~$ sudo gpasswd -d qa1 quality
Removing user qa1 from group quality
lubuntu@lubuntu:~$ id qa1
uid=1003(qa1) gid=1003(qa1) groups=1003(qa1)

```

9. Create a user `lead1` whose primary group is `developers` and supplementary groups are `quality` and `support`.

```

lubuntu@lubuntu:~$ sudo groupadd developers
lubuntu@lubuntu:~$ sudo useradd lead1
lubuntu@lubuntu:~$ sudo usermod -g developers lead1
lubuntu@lubuntu:~$ sudo usermod -aG quality lead1
lubuntu@lubuntu:~$ sudo usermod -aG support lead1
lubuntu@lubuntu:~$ id lead1
uid=1801(lead1) gid=2501(developers) groups=2501(developers),1006(quality),2500(support)

```

10. List all users who belong to the `developers` group using `/etc/group`.

```

lubuntu@lubuntu:~$ cat /etc/group
developers:x:2501:

```

11. Create a user `tempuser` with expiry date after 30 days and verify.

```

lubuntu@lubuntu:~$ sudo useradd -e 2026-10-09 tempuser

```

12. Lock the account of `dev1` and then unlock it.

```
lubuntu@lubuntu:~$ sudo passwd -l dev1
passwd: password changed.
lubuntu@lubuntu:~$ sudo passwd -S dev1
dev1 L 2026-09-09 0 99999 7 -1
lubuntu@lubuntu:~$ sudo passwd -u dev1
passwd: password changed.
lubuntu@lubuntu:~$ sudo passwd -s dev1
^C lubuntu@lubuntu:~$ sudo passwd -S dev1
dev1 P 2026-09-09 0 99999 7 -1
```

13. Delete user `qa3` but keep its home directory.

```
lubuntu@lubuntu:~$ sudo userdel qa3
lubuntu@lubuntu:~$ sudo ls -l /home/qa3
total 0
```

14. Delete user `tempuser` along with its home directory.

```
lubuntu@lubuntu:~$ sudo userdel -r tempuser1
userdel: tempuser1 mail spool (/var/mail/tempuser1) not found
lubuntu@lubuntu:~$ ls -ld /home/tempuser1
ls: cannot access '/home/tempuser1': No such file or directory
lubuntu@lubuntu:~$
```

B. File / Directory Permissions - Basic chmod

15. Create a directory `app1` and create 3 files `file1.txt`, `file2.txt`, `file3.txt` inside it.

```
lubuntu@lubuntu:~$ mkdir app1
lubuntu@lubuntu:~$ touch app1/{file1.txt,file2.txt,file3.txt}
lubuntu@lubuntu:~$ tree app1
app1
├── file1.txt
├── file2.txt
└── file3.txt
```

16. Check default permissions of file and directory and explain umask effect.

```
touch app1/file1.txt app1/file2.txt app1/file3.txt
pgcp-esd@pc:~$ umask
ls -ld app1
ls -l app1/file1.txt
0002
drwxrwxr-x 2 pgcp-esd pgcp-esd 4096 Sep 10 22:47 app1
-rw-rw-r-- 1 pgcp-esd pgcp-esd 0 Sep 10 22:47 app1/file1.txt
```

17. Create a file `secret.txt` and set permissions to `rwx-----` using octal.

```
lubuntu@lubuntu:~/app1$ touch secret.txt
lubuntu@lubuntu:~/app1$ chmod 700 secret.txt
lubuntu@lubuntu:~/app1$ ls -l secret.txt
-rwx----- 1 lubuntu lubuntu 0 Sep  9 05:35 secret.txt
```

18. Create a file `public.txt` and set permissions to `r--r--r--` using symbolic notation.

```
lubuntu@lubuntu:~/app1$ chmod a-rwx,a+r public.txt
lubuntu@lubuntu:~/app1$ ls -l public.txt
-r--r--r-- 1 lubuntu lubuntu 0 Sep  9 05:37 public.txt
```

19. Set permissions `750` on directory `app1` and test if others can `cd` into it.

```
lubuntu@lubuntu:~$ chmod 750 app1
lubuntu@lubuntu:~$ su dev1
Password:
dev1@lubuntu:/home/lubuntu$ ls
ls: cannot open file '.': Permission denied
```

20. Set permissions on `file1.txt` to `owner=rwx, group=r--, other=---` using both symbolic and octal.

```
lubuntu@lubuntu:~/app1$ setfacl -m u:lubuntu:rwx file1.txt
lubuntu@lubuntu:~/app1$ setfacl -m g::r-- file1.txt
lubuntu@lubuntu:~/app1$ setfacl -m o::--- file1.txt
```

21. Add write permission for group on `file2.txt` without disturbing other permissions.

```
lubuntu@lubuntu:~/app1$ chmod g+w file2.txt
```

22. Remove execute permission for others from `app1` directory.

```
lubuntu@lubuntu:~$ chmod o-x app1
lubuntu@lubuntu:~$ ls -l app1
total 0
-rw-rwx---+1 lubuntu lubuntu 0 Sep  9 05:29 file1.txt
-rw-rw-r-- 1 lubuntu lubuntu 0 Sep  9 05:29 file2.txt
-rw-rw-r-- 1 lubuntu lubuntu 0 Sep  9 05:29 file3.txt
-r--r--r-- 1 lubuntu lubuntu 0 Sep  9 05:37 public.txt
-rw-r--r-- 1 lubuntu lubuntu 0 Sep  9 05:39 public1.txt
-rwx----- 1 lubuntu lubuntu 0 Sep  9 05:35 secret.txt
```

23. Set recursive permissions `770` on `app1` directory and all files inside it.

```

ubuntu@ubuntu:~$ chmod -R 770 app1
ubuntu@ubuntu:~$ ls -l app1
total 0
-rwxrwx---+1 ubuntu ubuntu 0 Sep  9 05:29 file1.txt
-rwxrwx--- 1 ubuntu ubuntu 0 Sep  9 05:29 file2.txt
-rwxrwx--- 1 ubuntu ubuntu 0 Sep  9 05:29 file3.txt
-rwxrwx--- 1 ubuntu ubuntu 0 Sep  9 05:37 public.txt
-rwxrwx--- 1 ubuntu ubuntu 0 Sep  9 05:39 public1.txt
-rwxrwx--- 1 ubuntu ubuntu 0 Sep  9 05:35 secret.txt

```

24. Create a file `data.log` with `660` and test access with owner, group member, and other user.

```

pgcp-esd@pc:~$ chmod -R 770 app1
pgcp-esd@pc:~$ touch data.log
chmod 660 data.log
# Owner and group members can read/write; others get permission denied.
pgcp-esd@pc:~$

```

C. Ownership, Special Permissions, Sticky Bit & SGID

25. Change owner of `app1/file1.txt` to `dev1` and group to `developers`.

```

pgcp-esd@pc:~$ sudo chown dev1:developers app1/file1.txt

```

26. Change group ownership of entire `app1` directory recursively to `developers`.

```

pgcp-esd@pc:~$ sudo chgrp -R developers app1

```

27. Create a directory `collab` with permissions `2770` and explain the purpose of SGID.

```

pgcp-esd@pc:~$ mkdir collab
chmod 2770 collab

```

SGID forces all files and subdirectories created within `collab` to inherit the group ownership of `collab` rather than the creator's primary group.

28. Create a directory `tmpwork` with permissions `1777` and test sticky bit behavior - create files as `dev1` and `dev2` and try to delete each other's file.

```
pgcp-esd@pc:~$ mkdir tmpwork
chmod 1777 tmpwork
```

29. Create a file with SUID bit set and observe permission string.

```
# users can create files, but can only delete files they
pgcp-esd@pc:~$ touch suid_file
chmod u+s suid_file
ls -l suid_file
-rwsr--r-- 1 pgcp-esd pgcp-esd 0 Sep 10 22:57 suid_file
```

30. Remove SUID, SGID and Sticky bit from all files in a directory.

```
pgcp-esd@pc:~$ chmod -R u-s,g-s,o-t directory/
chmod: cannot access 'directory/': No such file or directory
pgcp-esd@pc:~$
```

31. Create a directory `private` such that only `dev1` can access it, group and others have no rights.

```
pgcp-esd@pc:~$ mkdir private
sudo chown dev1 private
chmod 700 private
chmod: changing permissions of 'private': Operation not permitted
```

32. Create a directory `readonly` where owner can do anything, group can only list files but not create, others no access.

```
pgcp-esd@pc:~$ mkdir readonly
chmod 750 readonly
pgcp-esd@pc:~$
```

D. Advanced ACL Practice

33. Create a file `project\_plan.txt` and display its ACL with `getfacl`.

```
lubuntu@lubuntu:~$ mkdir project
lubuntu@lubuntu:~$ touch project/project_plan.txt
lubuntu@lubuntu:~$ getfacl project/project_plan.txt
# file: project/project_plan.txt
# owner: lubuntu
# group: lubuntu
user::rw-
group::r--
other::r--
```

34. Give `dev1` rwx, `dev2` r--, `dev3` --- access on `project\_plan.txt` using `setfacl` and verify.

```
lubuntu@lubuntu:~/project$ setfacl -m u:dev1:rwx project_plan.txt
lubuntu@lubuntu:~/project$ setfacl -m u:dev2:r-- project_plan.txt
lubuntu@lubuntu:~/project$ setfacl -m u:dev3:--- project_plan.txt
lubuntu@lubuntu:~/project$ getfacl project_plan.txt
# file: project_plan.txt
# owner: lubuntu
# group: lubuntu
user::rw-
user:dev2:r--
user:dev1:rwx
user:dev3:---
group::r--
mask::rwx
other::r--
```

35. Remove ACL entry of `dev2` from `project\_plan.txt`.

```
lubuntu@lubuntu:~/project$ setfacl -x dev2 project_plan.txt
lubuntu@lubuntu:~/project$ getfacl project_plan.txt
# file: project_plan.txt
# owner: lubuntu
# group: lubuntu
user::rw-
user:dev1:rwx
user:dev3:---
group::r--
mask::rwx
```

36. Create a directory `acl\_project` and set default ACL so that all new files inside automatically get `r--` for `dev2`.

```
lubuntu@lubuntu:~/project$ mkdir acl_project
lubuntu@lubuntu:~/project$ setfacl -m u:dev2:r-- acl_project
lubuntu@lubuntu:~/project$ getfacl acl_project
# file: acl_project
# owner: lubuntu
# group: lubuntu
user::rwx
user:dev2:r--
group::r-x
mask::r-x
other::r-x
```

37. Give group `testers` rw- permission on `acl\_project` using ACL.

```
lubuntu@lubuntu:~/project$ sudo groupadd testers
lubuntu@lubuntu:~/project$ setfacl -m g:testers:rw- acl_project
lubuntu@lubuntu:~/project$ getfacl acl_project
# file: acl_project
# owner: lubuntu
# group: lubuntu
user::rwx
user:dev2:r--
group::r-x
group:testers:rw-
mask::rwx
other::r-x
```

38. Remove all ACL entries from a file and restore to normal Unix permissions.

```
lubuntu@lubuntu:~/project$ setfacl -b project_plan.txt
lubuntu@lubuntu:~/project$ getfacl project_plan.txt
# file: project_plan.txt
# owner: lubuntu
# group: lubuntu
user::rw-
group::r--
other::r--
```

39. Create a scenario where permission is `700` but `dev2` can still read via ACL.

```
lubuntu@lubuntu:~/project$ touch file4.txt
lubuntu@lubuntu:~/project$ chmod 700 file4.txt
lubuntu@lubuntu:~/project$ setfacl -m u:dev2:r-- file4.txt
lubuntu@lubuntu:~/project$ getfacl file4.txt
# file: file4.txt
# owner: lubuntu
# group: lubuntu
user::rwx
user:dev2:r--
group:---
mask:r--
other:---
```

40. Use `getfacl` and `setfacl --restore` to backup and restore ACLs.

```
pgcp-esd@pc:~$ getfacl -R acl_project > acl_backup.txt
setfacl --restore=acl_backup.txt
```

E. sudo, su, and Access Testing

41. Check which commands current user can run with sudo using `sudo -l`.

```
lubuntu@lubuntu:~/project/acl_project$ sudo -l
User lubuntu may run the following commands on lubuntu:
  (ALL : ALL) ALL
  (ALL) NOPASSWD: ALL
```

42. Run `whoami` and `id` normally, then with `sudo` and compare outputs.

```
lubuntu@lubuntu:~/project/acl_project$ whoami
lubuntu
lubuntu@lubuntu:~/project/acl_project$ id
uid=1000(lubuntu) gid=1000(lubuntu) groups=1000(lubuntu),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),110(lpadmin),115(sambashare)
lubuntu@lubuntu:~/project/acl_project$ sudo whoami
root
lubuntu@lubuntu:~/project/acl_project$ sudo id
uid=0(root) gid=0(root) groups=0(root)
```

43. Create a user `junior\_admin` and allow him to run only `useradd` command via sudo without password.

```
pgcp-esd@pc:~$ sudo useradd -m -s /bin/bash 'junior_admin'
[sudo] password for pgcp-esd:
pgcp-esd@pc:~$ cat /etc/passwd | tail -1
junior_admin:x:1018:1026::/home/junior_admin:/bin/bash
pgcp-esd@pc:~$ sudo passwd junior_admin
New password:
Retype new password:
passwd: password updated successfully
pgcp-esd@pc:~$ sudo visudo -f /etc/sudoers.d/junior_admin
pgcp-esd@pc:~$
```

```
junior_admin ALL=(ALL) NOPASSWD: /usr/sbin/useradd
```

```
pgcp-esd@pc:~$ su junior_admin
Password:
junior_admin@pc:/home/pgcp-esd$ sudo useradd test_user
junior_admin@pc:/home/pgcp-esd$ sudo apt update
[sudo] password for junior_admin:
sudo: a password is required
junior_admin@pc:/home/pgcp-esd$
```

44. Configure sudoers so that `admins` group can run all commands as root.

```
pgcp-esd@pc:~$ sudo visudo
visudo: /etc/sudoers.tmp unchanged
pgcp-esd@pc:~$ sudo chmod 0440 /etc/sudoers.d/admins
pgcp-esd@pc:~$ sudo -l
Matching Defaults entries for pgcp-esd on pc:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty

User pgcp-esd may run the following commands on pc:
    (ALL : ALL) ALL
```

```
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL
%admins ALL=(ALL:ALL) ALL
```

45. Switch to `dev1` using `su -` and `su` and show the difference in environment variables `pwd` and `HOME`.

```
pgcp-esd@pc:~$ sudo visudo
pgcp-esd@pc:~$ su - dev1
Password:
dev1@pc:~$ whoami
dev1
dev1@pc:~$ exit
logout
pgcp-esd@pc:~$ su dev1
Password:
dev1@pc:/home/pgcp-esd$ whoami
dev1
dev1@pc:/home/pgcp-esd$ exit
exit
pgcp-esd@pc:~$
```

46. As `dev1`, create a file `dev1\_report.txt` with `640`, login as `dev2` [same group] and as `student1` and test read/write.

```
dev1@pc:/home/pgcp-esd$ cd ~
dev1@pc:~$ echo "content" > dev1_report.txt
dev1@pc:~$ chmod 640 dev1_report.txt
dev1@pc:~$ su dev2
Password:
dev2@pc:/home/dev1$ nano dev1_report.txt
dev2@pc:/home/dev1$
dev2@pc:/home/dev1$ su student1
Password:
student1@pc:/home/dev1$ nano dev1_report.txt
```

```
Error writing dev1_report.txt: Perm
^G Help      ^O Write Ou^W Where Is
^X Exit      ^R Read Fil^_ Replace
```

```
dev1_report.txt

[ Path '.:': Permission denied ]
^G Help      ^O Write Ou^W Where Is
^X Exit      ^R Read Fil^_ Replace
```

47. As root, create a file `root\_file.txt` with `600` in `/tmp` and check if `manager` can read it.

```
root@pc:/home/pgcp-esd# touch /tmp/root_file.txt
root@pc:/home/pgcp-esd# chmod 600 /tmp/root_file.txt
root@pc:/home/pgcp-esd# echo "Created by root" > /tmp/root_file.txt
root@pc:/home/pgcp-esd# exit
exit
pgcp-esd@pc:~$ su manager
Password:
manager@pc:/home/pgcp-esd$ cd #
manager@pc:~$ ls
manager@pc:~$ cat /tmp/root_file.txt
cat: /tmp/root file.txt: Permission denied
```

48. Execute a single command `cat /etc/shadow` as root using `sudo` without switching to root shell.

```
pgcp-esd@pc:~$ sudo cat /etc/shadow
root:!:20684:0:99999:7:::
daemon*:19576:0:99999:7:::
bin*:19576:0:99999:7:::
sys*:19576:0:99999:7:::
```

49. Create a user `auditor` with nologin shell and verify login is denied.

```
pgcp-esd@pc:~$ sudo useradd -s /usr/sbin/nologin auditor
pgcp-esd@pc:~$ sudo passwd auditor
New password:
Retype new password:
Sorry, passwords do not match.
New password:
Retype new password:
passwd: password updated successfully
pgcp-esd@pc:~$ grep auditor /etc/passwd
auditor:x:1020:1028::/home/auditor:/usr/sbin/nologin
pgcp-esd@pc:~$ su auditor
Password:
This account is currently not available.
pgcp-esd@pc:~$
```

50. Create a custom home directory `/opt/intern\_work` for user `intern2` and set permissions so only owner and `developers` group can access it.[other]

```
pgcp-esd@pc:~$ sudo useradd -m -d /opt/intern_work intern2
pgcp-esd@pc:~$ sudo passwd intern2
New password:
Retype new password:
passwd: password updated successfully
pgcp-esd@pc:~$ grep intern2 /etc/passwd
intern2:x:1021:1029::/opt/intern_work:/bin/sh
pgcp-esd@pc:~$ sudo chown -R intern2:developers /opt/intern_work
pgcp-esd@pc:~$ sudo chmod 770 /opt/intern_work
pgcp-esd@pc:~$ ls -ld /opt/intern_work
drwxrwx--- 2 intern2 developers 4096 Sep 10 22:32 /opt/intern_work
pgcp-esd@pc:~$
```